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The Multiphase Flow Dynamic Simulator Application for Optimization for Well Performance

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Abstract

Multiphase flow modeling in wells is an important component of oil and gas field development, where accurate prediction of fluid dynamics in transient regimes directly affects production efficiency and operational safety. This paper presents Flow Simulator 1D (FS1D), a one-dimensional drift-flux-based numerical simulator designed for dynamic modeling of multiphase flow in wells and pipes. The simulator solves problems associated with unsteady multiphase behavior by solving the equations of conservation of mass, momentum and energy in a fully implicit formulation, allowing multiphase flows to be modeled under various flow regimes and well inclinations. FS1D integrates pre- and post-processing into a single platform and supports complex boundary conditions, inflow modeling, reservoir heat transfer, and modeling of various equipment such as electric submersible pumps (ESPs) and chokes. The use of a drift-flux flow model improves computational efficiency while maintaining the necessary accuracy for engineering analysis, especially in evaluating flow distribution, pressure drops, phase segregation, and thermal effects. The simulator has been validated on synthetic cases that reproduce field scenarios including transient flow dynamics, gas entry, gravity segregation, and non-isothermal flow behavior. FS1D's modular architecture and restart capabilities also make it a powerful tool for creating synthetic datasets and supporting digital oilfield applications.

Introduction

Multiphase flow, characterized by the simultaneous movement of oil, water, and gas, plays a key role in various scenarios of hydrocarbon extraction and transportation. Phase transitions, changes in fluid properties, and determination of their behavior under different pressure and temperature conditions are important for its analysis. One of the applied tasks is the flow in the wellbore, where accurate prediction of multiphase dynamics is necessary to increase efficiency and monitor the development of oil and gas fields. The use of multiphase fluid flow simulators in the wellbore to refine transient processes is an important part of reducing uncertainties in well operation management.

The accuracy of modeling transient processes in the wellbore is important for making decisions regarding well management. Such tasks involve modeling with time intervals ranging from several hours to several

weeks. This aspect imposes constraints on the choice of mathematical models and numerical methods that can be effectively used in the corresponding multiphase fluid flow simulators. Typically, in such cases, multiphase flow modeling involves multi-fluid models (K. H. Bendiksen et al. 1991), drift-flux models (N. Zuber et al. 1965), Baer-Nunziato-type models (M.R. Baer et al. 1986), and simplified steady-state models (M. Jerez-Carrizales et al. 2015).

One of the key principles of well operation is determining the flow rate of each fluid type in a multiphase flow. Accurate measurements of the flow rates of different phases, their properties, and dynamic characteristics are an integral part of optimal well management. Insufficient data accuracy can lead to errors in forecasts, which in turn affects decision-making regarding well operation optimization. The influence of multiphase flow uncertainties on well operation parameters necessitates the development of methods to reduce them. Research in this direction focuses on developing adequate mathematical models that consider various factors such as geological features, fluid characteristics, fluid flow physics, and operating conditions. Consequently, multiphase fluid flow simulators become important tools for improving management processes, providing the ability to conduct virtual experiments and optimize operation parameters well. They also account for complex phase interactions, minimize uncertainties, and enhance forecast accuracy, thereby contributing to well optimization and increasing the economic efficiency of production.

This paper presents a new simulator of dynamic one-dimensional multiphase liquid flow – Flow Simulator 1D (FS1D) (A. Gryzlov et al. 2023), developed specifically for the management and optimization of oil and gas well operations. Unlike existing commercial programs, this simulator utilizes a drift-flux flow model, providing advantages in reducing complexity and improved tuning capabilities using field data.

Modeling multiphase flow in the FS1D simulator

Multiphase flow modeling has long established itself as a reliable and practical method for quantitatively analyzing the behavior of flows in wells, flow lines and pipelines. The essence of the approach lies in the numerical solution of systems of equations for the conservation of mass, momentum and energy, describing the motion of multiphase media under conditions close to real cases. Such modeling allows not only to take into account the complex thermo-hydrodynamics of fluids, but also to accurately reproduce transient flow regimes - from laminar to turbulent, from bubble to annular. Such models are especially valuable in solving applied problems of field development. For example, when planning a well kill under conditions of high reservoir pressures and temperatures, modeling allows to estimate the dynamics of gas influx and calculate the parameters of cement slurries that ensure safe restoration of control over the well. Similarly, when drilling in zones prone to absorption, flow modeling becomes a core of the managed pressure distribution strategy, allowing to balance the densities of the drilling fluid and reduce the risk. Flow models are also actively used to predict emergency blowout scenarios. They allow to estimate the potential flow rate of a well in the event of a leak and determine the parameters for pumping kill fluids - volumes, densities, speeds - required to stabilize the situation. In addition, based on the modeling results, algorithms for action in the event of an emergency are developed, including the sequence of operations, permissible operating modes and maximum operating parameters of equipment. In industrial practice, flow modeling helps to determine the contribution of individual productive reservoirs to the total flow rate, optimize well operating performances, prepare well strategy to change the flow rate or conduct a safe start-up/stop of the pipeline.

The significance of applying one-dimensional numerical modeling of multiphase flow in wellbores is due to the large aspect ratio of well lengths to diameters. The formulation of the simulator is based on the temporal and spatial averaging of the Navier-Stokes equations, with a focus on the drift-flux approach, making it less reliant on closure relations. Mass conservation equations are discretized and numerically solved using stable algorithms, ensuring the model's applicability to a wide range of tasks in the oil and gas industry. Despite its simplicity compared to multi-fluid models, the drift-flux model plays a key role

in improving computational performance by reducing the number of equations required for solution. This approach maintains the necessary modeling accuracy while achieving acceptable computation times. The diagram below in Figure 1 illustrates the approach to identifying flow at each grid point in the modeling scheme. For a fully developed flow and stabilized regimes, such a one-dimensional formulation is obtained from averaging the Navier-Stokes equations over cross-section and over time with the introduction of additional closing relations depending on the flow regime. Diffusion in the one-dimensional model is usually neglected, and the flow is considered as completely convective. The resulting simplified equations are discretized on a finite number of elements of the computational grid, after which they are solved numerically.

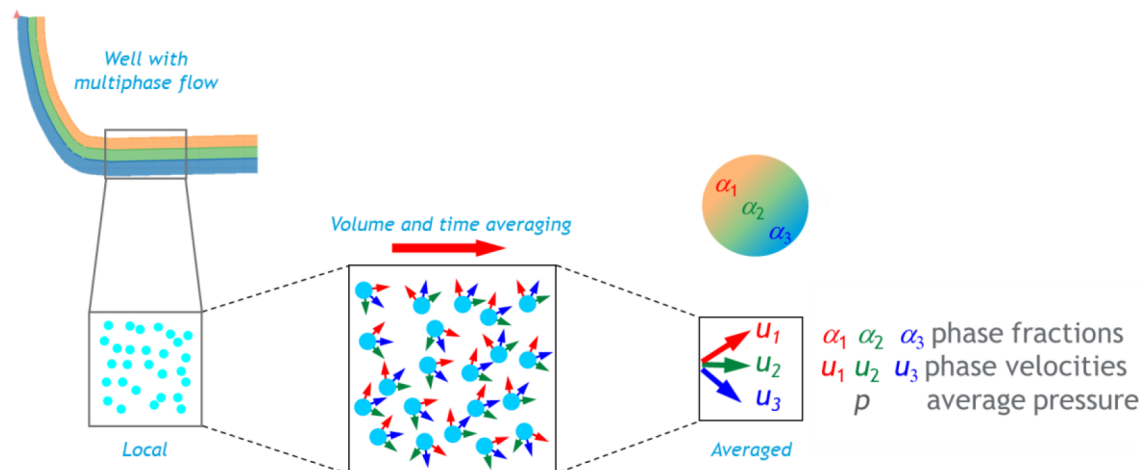


Figure 1—One-dimensional model (1D) scheme for calculating multiphase flow in the wellbore.

The developed multiphase flow simulator FS1D adheres to a widely adopted industry approach in which the entire physics of multiphase flow is approximated by a set of one-dimensional differential equations combined with empirical correlations. The fluid flow domain, represented by the wellbore, is discretized into a grid of non-intersecting cells, and for each of these cells, the corresponding dynamic variables are numerically calculated.

The basic principle of the simulator is based on software logic for modeling, where the entire process is divided into three stages. The first stage is preprocessing, in which the user creates a simulation model with the necessary parameters and initial data. In the second stage, the prepared model is then passed to the solver, where the main equations are numerically solved, and the distribution of dynamic variables is stored locally. The final stage is the postprocessor, where the obtained results can be visualized in the form of graphs or saved in a suitable format. The workflow and interface of the simulator are depicted in the diagram below in Figure 2.



Figure 2—Interface of the developed well simulator

Main features of the simulator:

- Fully integrated pre- and post-processing within a single software product;
- Modeling transient processes in a single well of all possible inclination;
- Fully implicit numerical solver;
- Three-phase fluid flow accounting for simultaneous gas, oil, and water flow and the volume fraction of each phase;
- The model considers separate velocities for each fluid, with the difference between these velocities modeled by the drift-flux equation;
- Gas segregation model implemented according to the black oil model, allowing for the simulation of equilibrium gas segregation and dissolution of natural gas into/from the oil phase;
- Ability to account for inflow from the reservoir into the wellbore using various analytical methods;
- Ability to account for the operation of electric submersible pumps (ESP) and chokes;
- Option for restarting calculations from a specific time step with the ability to modify input data.

Figure 3 shows the main interface of the simulator. On the right, users can access a list of tutorials designed to familiarize themselves with the software. The name of the active project is displayed at the top of the screen. The left panel consists of three main sections: (input) model setup, (solver) simulation execution, and (results) results visualization.

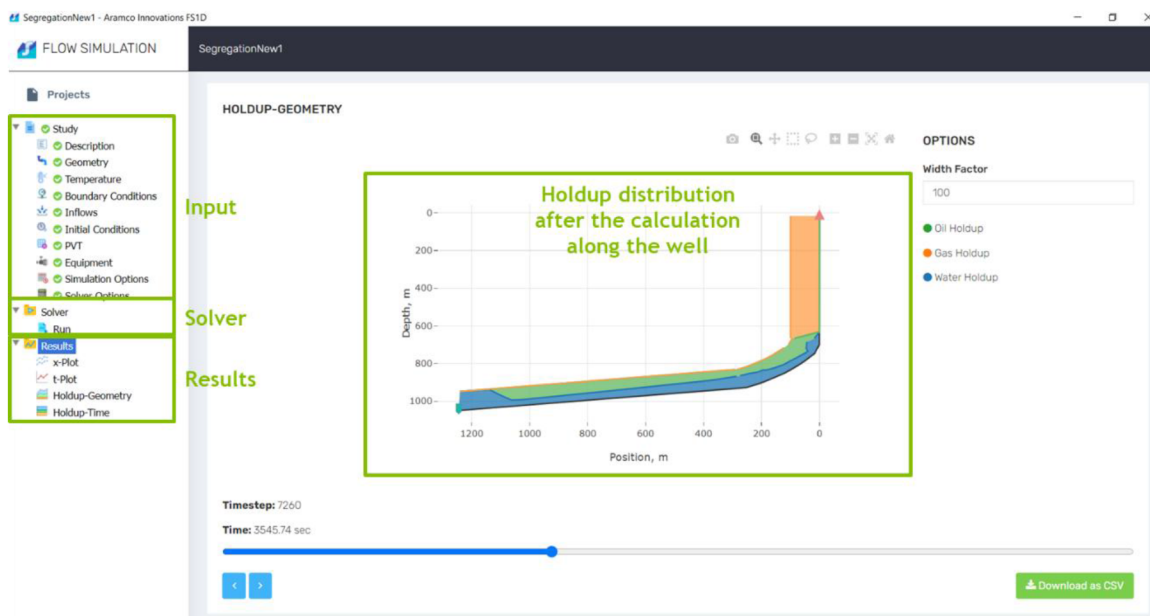


Figure 3—Interface of the developed the FS1D simulator

In the initial preprocessing stage, the user defines the wellbore geometry by defining segments of a given length. Then, boundary and initial conditions must be defined, and models for inflow from reservoir must be selected. Fluid properties, typically obtained from laboratory experiments or other numerical models, are entered in a tabular format to reflect realistic hydrocarbon behavior. The simulator also allows the use of downhole equipment such as electric submersible pumps or chokes. The solver panel allows users to stop or restart the simulation. The restart function is particularly useful for continuing the simulation from a selected point in time, either with the same settings or with modified parameters, which is especially valuable when creating large synthetic data sets under changing flow conditions. An advanced feature of the simulator is

the holdup diagram, which displays the volumetric phase distribution along the wellbore. A time step slider allows users to dynamically explore the fluid distribution along the wellbore trajectory.

Demonstration of the FS1D on synthetic cases

Flow rate estimation

This example on the [Figure 4](#) presents a detailed calculation of flow rate distribution along the wellbore using FS1D, which is designed to address the complex aspects of multiphase flow modeling used in the industry where multiple phases such as oil, water and gas interact simultaneously. FS1D is based on the solution of the conservation laws of mass, momentum and energy for each component phase, which allows accurate consideration of dynamic changes in fluid parameters under given flow conditions. The main input parameter for modeling is reservoir productivity, which acts as a boundary condition for fluid inflow from the reservoir. This approach allows to set realistic inflow parameters and ensure correct distribution of incoming flows over the nodes of the modeled wellbore, which subsequently becomes the basis for calculating the average velocities of each phase. This approach allows not only setting realistic inflow parameters, but also discretizing the wellbore into elementary sections, where local parameters - pressure, temperature, velocity of each phase are calculated, which ensures taking into account the influence of viscosity, density, capillary and gravity effects that significantly affect the flow dynamics. Based on the average velocity calculations for each phase component, the simulator determines the flow distribution along the wellbore, identifying areas with the highest hydraulic resistance, resulting in a decreasing trend in fluid velocity along the wellbore, which leads to a decrease in local fluid flow rate, as shown in the graph on the right side of the [Figure 4](#). The results demonstrate that the flow velocity along the wellbore tends to decrease when moving away from the inflow zone. This phenomenon is caused by a combination of factors: increase in hydraulic resistance due to friction between phases, the effect of gravitational forces, and a change in the phase ratio as a result of phase separation. Decrease of local velocity leads not only to decrease of liquid flow rate, but also to change of hydrodynamic regime, which can be accompanied by transition from bubble flow regime to jet flow regime or even to drainage regime. These transitions are critical to accurately predicting well behavior because they can significantly affect production efficiency and pressure distribution along the wellbore. These calculated data not only enable more accurate prediction of well behavior under different operating conditions, but also optimize production by adjusting operating regimes, changing pumping equipment or redesigning process regimes to help reduce the risk of inefficient production and increase economic returns from field development.



Figure 4—Calculation of flow rate in the FS1D

Modeling Transition of Gas from Liquid in the Wellbore

This study (Figure 5) evaluates a gas entry scenario that occurs when a well encounters a transition from liquid-dominated to gas-dominated flow due to the gas lift or a change in the fluid flow conditions. The numerical modeling of this process is performed using a multiphase simulator capable of solving the mass, momentum and energy conservation equations for each phase component to account for the complex interplay between the gas and gas flowing through the wellbore. The initial pressure distribution along the wellbore shows a uniform pressure increase with depth, which is typical for the hydrostatic regime, but the entry of gas at a depth of about 500 m leads to a noticeable change in the pressure profile. The graph (Figure 5) goes from a linear increase to a curved curve reflecting a local pressure decrease due to the interaction of low-density gas flow with denser fluids, which indicates the beginning of phase separation. The plots on the right (Figure 5) show the evolution of the phase distribution within the wellbore, initially, the predominant oil-water ratio changes in the gas breakthrough zone, and in the final step of the calculation, the simulator captures the overall transition to a multiphase configuration where gas flow has a significant influence on the flow dynamics, altering the production regime and creating new hydrodynamic scenarios. This integrated simulation approach not only provides an estimate of entered gas, complementing the well test results, but also provides critical information for further analysis, development of preventive measures and optimization of operational well control strategies, which has a direct impact on the reliability and cost-effectiveness of well performance under changing performances scenarios. These calculations enable the estimation of amount of gas and provide data that, combined with well tests, can aid in planning a set of actions to address the situation. The obtained information helps in selecting the optimal well operation mode to minimize the impact of gas influx on production.

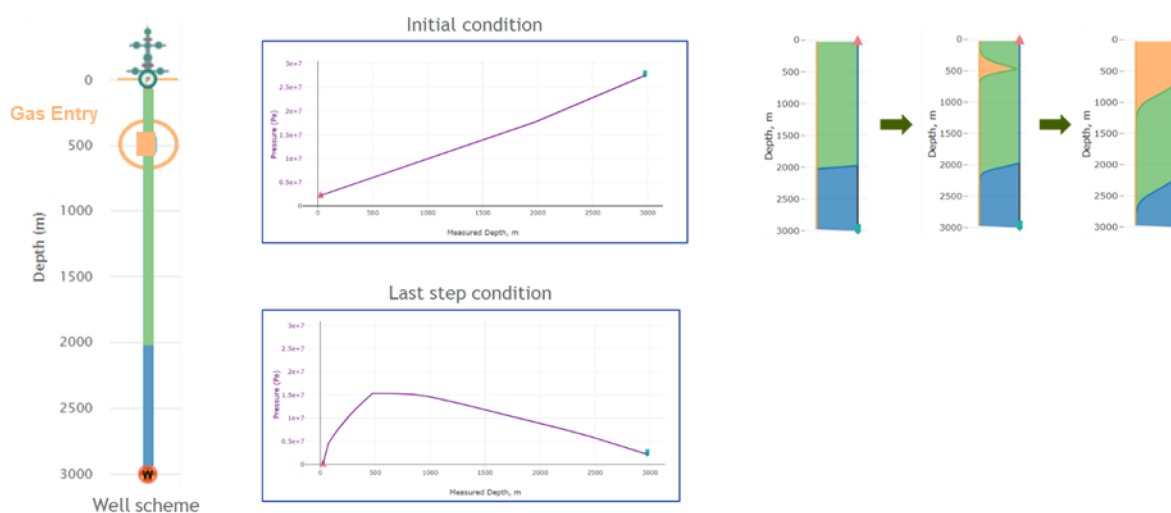


Figure 5—Gas transition simulation in the FS1D from the casing to the tubing

Three-Phase Flow Modeling with Variable Phase Segregation

In this example (Figure 6), the behavior of a three-phase system including gas, oil and water in a closed vertical well is considered. The initial phase distribution is physically unstable due to artificial mixing of the components and lack of gravitational equilibrium. This approach allows to investigate the natural processes of gravitational segregation that occur when the external influence ceases, which is critical for understanding the initial stages of formation of stable phase boundaries.

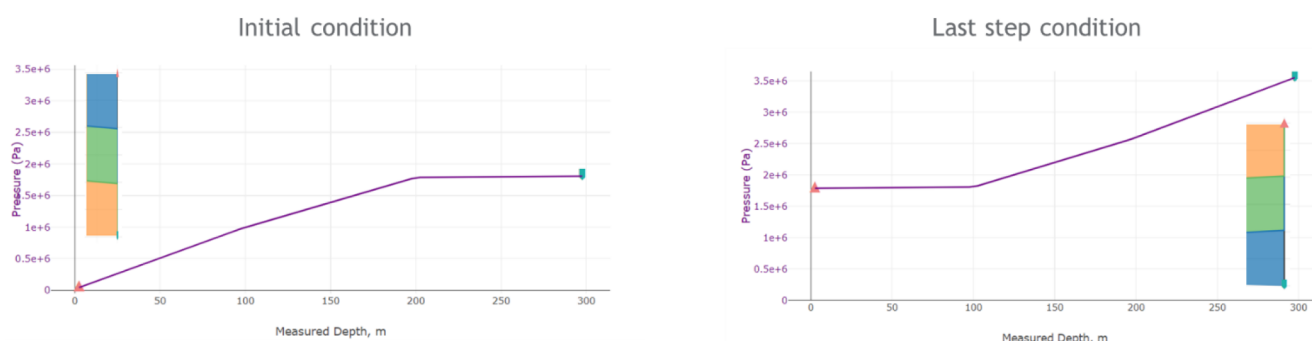


Figure 6—Three-Phase modeling complicated by varying segregation rates in FS1D

Gravitational forces play a key role in the redistribution of phases along the wellbore, leading to a gradual equalization of the density profile and the formation of characteristic zones of gas accumulation in the upper part of the wellbore and water accumulation in the lower part. The plots (Figure 6) show the pressure and phase saturation distributions at the initial and final calculation steps. The initial state is characterized by nearly linear pressure and chaotic phase distribution. As time progresses, phase segregation occurs, and as a result, the gas phase is concentrated in the upper part of the well, while the liquid phases are displaced downward. This leads to a marked change in the pressure profile, which becomes non-linear and reflects steady gravity stratification.

The obtained modeling results allow to evaluate the effect of phase segregation on the hydrostatic pressure distribution, which is critical in drilling fluid design. Correct modeling of these processes helps to reduce the risks of complications such as gas saturated zones, cavern formation and circulation failure. Thus, this simulation could be an important tool for optimizing drilling and selecting weighted mud parameters, providing safer and more predictable operations in wells with complex phase regimes.

Validating Heat Transfer in Multiphase Wellbore Flow

In this case (Figure 7), the progress of the temperature profile along a vertical well under non-isothermal flow conditions is analyzed. The model takes into account both mass and momentum transfer and energy exchange between the fluid in the well and the surrounding reservoir. A constant pressure condition is set at the upper boundary of the well, while the lower boundary is a flowing section into which fluid with a reservoir temperature is injected. The reservoir temperature varies with depth according to a given geothermal gradient.

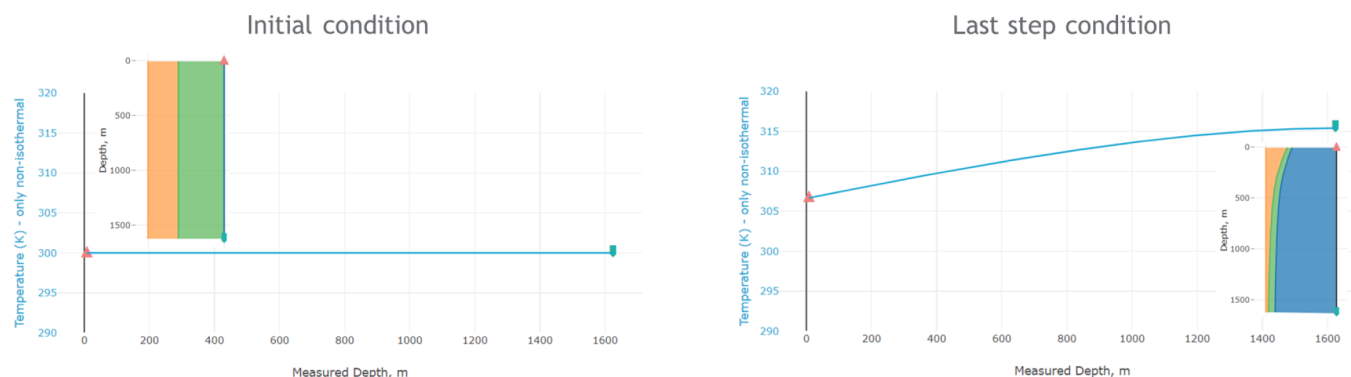


Figure 7—Simulation results of non-isothermal flow in the FS1D

The mathematical formulation of the problem involves solving the heat transfer equation associated with the equations of motion of a multiphase medium. The effects of convective and heat conduction transfer, as well as heat exchange with the surrounding reservoir through the well walls are taken into account. The

used model is realized on the basis of non-stationary approximation, which allows to accurately describe the process of temperature equilibrium establishment during fluid injection.

The graphs (Figure 7) show the temperature profiles at the initial and final stages of the calculation. The initial step is characterized by a uniform temperature in the well corresponding to the minimum value of the geothermal gradient. After the start of injection of fluid with higher temperature, a thermal front propagation up the well is observed. The final step shows a steady temperature profile that exceeds the temperature gradient of the surrounding formation along the entire length of the well, especially in the upper part of the well. This indicates the dominance of the thermal influence of the fluid over the natural geothermal field.

Such calculations have important applications in planning thermal enhanced oil recovery methods (e.g., steam-thermal and hot water injection) and in estimating heat losses in transportation systems. Prediction of temperature in real time allows to increase the accuracy of control over technological performances, prevent thermal expansion of equipment and ensure the safety of well operation at high temperatures. The use of non-isothermal modeling, taking into account heat transfer between fluid and rock, allows to reproduce more accurately the real thermodynamics of reservoir processes, increasing the reliability of numerical prediction in engineering practice.

Conclusions

Refinement of multiphase flow parameters under conditions of combined oil, water and gas flow is one of the key directions in flow control in wells and pipes in the field. Modeling of unsteady processes in the wellbore allows to significantly improve the accuracy of prediction, reduce uncertainties and make engineering decisions on well operation. However, prediction of multiphase flow behavior is complicated by high sensitivity to geological heterogeneities, variability of physical and chemical properties of fluids and unstable well operation performance.

Limited availability of measurements of phase flow rates and parameters inside the well creates the need to use numerical modeling as an additional tool to improve accuracy. The development and application of mathematical models that account for both reservoir and field fluid properties, as well as wellbore and equipment design features, is becoming a critical factor in data interpretation and operations planning.

The FS1D simulator is a dynamic (non-stationarity) multiphase flow model designed for numerical calculation of phase saturations, pressures, velocities, densities and temperatures along the well. The core of the simulator is based on a fully implicit numerical scheme that ensures stability and accuracy of solutions even under sharp transients. The use of the drift-flux equation models allows to take into account relative phase velocities, and the black oil model with equilibrium gas release allows to correctly describe the behavior of gas dissolved in oil and phase transitions.

FS1D can be effectively used both in stand-alone mode to analyze local conditions in the well and as part of complex models integrated with above-surface networks, providing a complete interface between the well model and the production system.

FS1D functions as a universal tool for engineering analysis, allowing to conduct virtual experiments, investigate sensitivity of process parameters, optimize operating performances and ensure reliable well control at all stages of field development. The combination of physical validity, numerical stability and engineering applicability makes FS1D a support tool for decision making under conditions of high uncertainty of the multiphase flow in wells.

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